Kaoutar Sarsari

PhD Candidate · AI & Databases · Teaching Assistant · Software Engineer

I study how machines learn, and I like teaching people the same thing.

LaSTI Laboratory, ENSA Khouribga, Sultan Moulay Slimane University, Morocco • kaoutar.sarsari.ensak@usms.ac.ma
sarsari1.github.io/sarsarikaoutar • linkedin.com/in/kaoutar-sarsari • github.com/SARSARI1 •

Research Profile

Second-year PhD researcher and teaching assistant at the National School of Applied Sciences (ENSA) of Khouribga, Sultan Moulay Slimane University, conducting research at the LaSTI Laboratory under the supervision of Prof. Nouredine Gherabi. My doctoral research lies at the intersection of **artificial intelligence, machine learning and databases**, and is supported by a CNRST scholarship. Trained as a software engineer, I combine theoretical rigour with a practical approach to building robust systems.

Research interests: Machine Learning | AI Research | Databases

Education

Dec 2025 – present
Ph.D. in Artificial Intelligence, Machine Learning & Databases
LaSTI Laboratory, ENSA Khouribga, Sultan Moulay Slimane University (USMS), Morocco
Supervisor: Prof. Nouredine Gherabi. Funding: CNRST doctoral scholarship. Thesis title: Application of Artificial Intelligence to Dynamic SQL Query Optimization and Adaptive Indexing

2020 – 2025
State Engineering Degree in Computer Science and Engineering
National School of Applied Sciences of Khouribga (ENSA Khouribga) – Honors ("Good")

2018 – 2020
Baccalaureate in Physical Sciences
Imam Ali Agouim High School, Ouarzazate – Highest Honors ("Very Good")

Publications

- K. Sarsari et al., "Evaluating Large Language Models for SQL Query Optimization: A Benchmark Study," International Conference on Convergent Technologies (ICCT 2026), Khouribga, Morocco, October 2026. Springer Lecture Notes in Networks and Systems (LNNS).

Teaching Experience

Dec 2025 – present
Teaching Assistant
ENSA Khouribga, Sultan Moulay Slimane University
Undergraduate courses in core computer science and AI: leading lab sessions, grading, and holding office hours. Courses taught: Operating Systems Theory, Unix, and Oracle Database Administration

Research & Professional Experience

Dec 2025 – present
PhD Researcher
LaSTI Laboratory, ENSA Khouribga
Doctoral research in AI and databases, in collaboration with academic and industry partners.

Feb 2025 – Sep 2025
Research Assistant, TPLTA Team
Oracle
Improved the internal TPLTA platform and managed advanced licensed datasets using APEX, PL/SQL, CSS, Liquibase, utPLSQL, OCI and XML.

Jul 2024 – Aug 2024
Software Engineering Intern
OCP Group – MyOCP application
Optimised and automated the MyOCP reservation-request workflow (Django, Bootstrap, JavaScript).

Aug 2023
Software Engineering Intern
OCP Group
Developed Phos-Predict-AI, a machine-learning application that calculates and predicts turnover (Flask, Bootstrap, JavaScript).

Invited Talks & Conference Roles

10 Dec 2025	**Speaker, TechnoBridge 2025** *"IT world: expectations vs reality"*
Jan 2026	**Roundtable Moderator, Open Source Days 14.0** *"Cloud in the Age of AI"*
22 Dec 2024	**Invited Speaker, Sawlat Event** *OCP Group Skills & Association Maher – seminar on Social Entrepreneurship*
Oct 2023 – Apr 2024	**Vice Chief of Conference, Open Source Days 12.0** *ENSA Khouribga* Co-led the conference committee: sourcing talks, workshops and guest speakers for the panels.
8–9 May 2023	**English Host, Open Source Days 11.0** *ENSA Khouribga* Presented the opening ceremony.

Leadership & Community Engagement

Nov 2024 – Feb 2025	**Team Leader, MDS Talks Season 2** *Moroccan Data Scientists (MDS)* Organised masterclass sessions with industry experts on tech trends and personal development, and led the organising team.
Apr 2024 – Jun 2024	**Graphic Designer (volunteer), MDS Talks** *Moroccan Data Scientists (MDS)* Created visual content and branding materials.
Oct 2023 – Apr 2024	**Deputy Campus Director, Hult Prize** *ENSA Khouribga* Supported campus operations and student entrepreneurship initiatives.

Entrepreneurship & Competitions

Jun – Sep 2022	**FRTN Technologies Startup Incubation Program** *Entrepreneurship training: startup creation, scaling and business acumen*
Apr 2022	**Odyssées Régionales (project "2ST")** *ENSAK – smart vehicle safety system to help prevent car accidents*
Dec 2020 – Apr 2021	**Hult Prize – 3rd place on campus** *Team Legacy Leavers, project "Aji l'Happiness Karma"; regional finals at the Bremen Impact Summit, April 2021*

Honours & Awards

- CNRST doctoral scholarship (2025 – present)
- First Place, Besaperk Competition, Khouribga, 2024
- Third place, Hult Prize on-campus programme, ENSA Khouribga (2021)
- Baccalaureate with Highest Honors (2020); State Engineering Degree with Honors (2025)

Skills

Programming	Python · Java · JavaScript · C · C++ · PHP · SQL · HTML/CSS
Frameworks	Symfony · Django · Flask · Spring Boot · Bootstrap · jQuery
Databases & Cloud	Oracle DB · PL/SQL · APEX · Liquibase · utPLSQL · OCI
Languages	Tamazight, Arabic, English, French